

Functions & the Haunted School

Post-Assessment 4 · 11 questions · 40 points

Name: _____

Date: _____

Score: / 40

This is a **post-assessment**. It shows what you learned in this unit. Try every question, and check your answers if you have time.

Work on your own, without a computer. You have about **45 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

1. Function vocabulary

2 points

```
def area(width, height):  
    return width * height  
  
print(area(3, 5))
```

Use a word from this bank: **parameter**, **argument**, **return value**, **docstring**.

(a) In this program, 3 and 5 are _____s. (1 pt)

(b) In this program, width and height are _____s. (1 pt)

2. Shout Machine

3 points

Write exactly what this program prints.

```
print("Hello")  
  
def shout(word):  
    print(word.upper() + "!!!")  
  
shout("pizza")  
shout("recess")  
print("Bye")
```

3. Banner Maker

4 points

```
def banner(text, symbol="*", width=3):  
    print(symbol * width + " " + text + " " + symbol * width)
```

Write what each call prints.

Call	Prints
<code>banner("Hi")</code>	
<code>banner("Win", "#")</code>	
<code>banner("Go", width=1)</code>	
<code>banner(symbol "=", text="End", width=2)</code>	

4. Return or print?

4 points

```
def add_tax(price):  
    return price + 5  
  
def show_tax(price):  
    print(price + 5)  
  
x = show_tax(10)  
y = add_tax(10)  
print(y)  
print(x)
```

(a) Write exactly what the program prints.

(3 pts)

(b) Why is `x` different from `y`?

(1 pt)

5. Is it a teen?

3 points

```
def is_teen(age):  
    return _____
```

- (a) Fill in the blank so `is_teen` returns `True` for ages 13 to 19 (including 13 and 19) and `False` otherwise. (2 pts)

- (b) With your function, what does `is_teen(20)` return? (1 pt)

6. Inside and outside

4 points

- (a) What does this print? (2 pts)

```
lives = 3  
  
def lose_life():  
    lives = 0  
    print("in function:", lives)  
  
lose_life()  
print("in game:", lives)
```

- (b) What happens when this runs, and why? (2 pts)

```
def roll():  
    result = 6  
  
roll()  
print(result)
```

7. Function Detective

3 points

```
def perimeter(width, height):  
    """Return the perimeter of a rectangle."""  
    return width + height * 2
```

(a) What is the line in triple quotes called?

(1 pt)

(b) A test says `perimeter(3, 5)` should be `16`. What does the function actually return?

(1 pt)

(c) Rewrite the `return` line so the test passes.

(1 pt)

8. First letters

3 points

```
def count_starts(words, letter):  
    total = 0  
    for w in words:  
        if w.startswith(letter):  
            total = total + 1  
    return total  
  
fruits = ["apple", "banana", "avocado", "cherry", "apricot"]  
print(count_starts(fruits, "a"))  
print(count_starts(fruits, "c"))  
print(count_starts(fruits, "z"))
```

Write exactly what the program prints.

9. Refactor Day

4 points

This code repeats itself. Write a function `notify(name, count)` that prints the line for one person, then write the calls that make the program print exactly the same thing.

```
print("Hi, Ana! You have 3 new messages.")
print("Hi, Ben! You have 5 new messages.")
print("Hi, Cy! You have 0 new messages.")
```


10. Acronym Maker

4 points

Write a function `acronym(phrase)` that **returns** the first letter of every word in capital letters. For example, `acronym("laugh out loud")` returns `"LOL"` and `acronym("be right back")` returns `"BRB"`.

11. Escape the Haunted School

6 points

Here is part of a text adventure.

```
rooms = {
    "cafeteria": {"west": "office", "up": "attic"},
    "office":    {"east": "cafeteria", "down": "basement"},
    "attic":     {"down": "cafeteria"},
    "basement": {"up": "office"},
}

def move(current, direction):
    if direction in rooms[current]:
        return rooms[current][direction]
    print("You can't go that way.")
    return current

room = "cafeteria"
for d in ["west", "west", "down", "up"]:
    room = move(room, d)
    print(room)
```

(a) Write exactly what the program prints.

(3 pts)

(b) Write a function `is_game_over(health, room)` that returns `True` when `health` is 0 or less **or** the player is in the `"basement"`, and `False` otherwise.

(3 pts)

END OF ASSESSMENT